

BizBook Marketing Module — Full Phased Roadmap (v2)

Architecture: one central Marketing Signal Engine. Every feature reads from it and writes back to it.

Legend: ● Standard | ● Advanced | ● Differentiator (India-specific moat)

Autonomy: ● Full Auto | ● Approve-Before-Send | ● Human-Only | — Not an action (analysis/reporting/UI)

Core principle: Marketing is the product. Everything routes back to one question the owner cares about: *where, when, and how much should I spend to grow revenue*. No feature below computes its own private version of "what's working" — everything reads from and writes to one shared brain.

Architecture — The Marketing Signal Engine

This is the single most important structural decision in the whole roadmap, and it's what your addition (Knowledge Graph + shared Signal Engine) correctly identifies as missing. Without it, the Experiment Lab, the Dynamic Pricing Optimizer, Journey Visualization, and the existing 8 opportunity-detection engines would each build their own version of "this customer/segment/channel is worth targeting" — and those versions will eventually disagree with each other, which destroys owner trust fast.

The fix: one engine, one source of truth.



What this means practically for Antigravity:

- The **Marketing Knowledge Graph** is the data model — nodes are products, customers, campaigns, festivals, inventory items, weather events, competitors, transactions; edges are relationships ("customer bought product during festival X," "campaign ran in locality Y during weather event Z"). This is what makes cross-feature insights possible — e.g., linking Dynamic Pricing to Local Area Intelligence to the Event Trigger Engine without three separate queries computing three separate answers.
- The **8 existing opportunity engines stay untouched in logic** — they become inputs to the graph, not separate silos.
- Every new engine (Experiment Lab, Dynamic Pricing Optimizer, Journey Visualization, Store Health Score) reads from the graph and writes its findings back into it, so the next engine downstream benefits automatically.
- **Confidence scoring lives at the engine level, not per-feature** — one calibrated confidence system, not five different ones with different meanings.

Phase 0 is now formally: build/extend the Signal Engine + Knowledge Graph schema before any of Phases 1–6 add features that would otherwise create duplicate logic.

Cross-Cutting System — Autonomy & Permission Layer

Autonomy Levels (applied per action type, not per feature):

-  **Full Auto** — system decides and executes with no human step.
-  **Approve-Before-Send** — system drafts/decides, owner gets a one-tap approve/reject.
-  **Human-Only** — system flags/recommends but never acts alone.

Required build components:

Component	What it does
Owner-Configurable Leash Settings	One-time setup: owner sets ₹ thresholds and category-level autonomy per action type.
Autonomy Registry	Config table mapping every action type to its current autonomy level per business — checked by the Co-Pilot before acting.
Trust-Based Leash Expansion	Uses the Track Record Scoreboard — after N successful auto actions with no override/complaint, system suggests raising the threshold.
Universal Action Log	Every action taken — auto or approved — gets logged and rolled into the Executive Daily Brief / weekly digest.

Launch conservative (favor  over ), loosen over time as trust builds.

Phase 0 — Signal Engine & Knowledge Graph Foundation (Prep, not a feature)

Status: AUDITED by Antigravity — real schema below replaces earlier placeholder assumptions.

Correction to earlier assumption: the system does not have "8 opportunity detection engines." It actually has **4 Opportunity Engines** (`marketingEngine.js`): Revenue Recovery, Customer Retention, Dead Stock Liquidation, Cross-Sell) + **4 Rule-Based Insight Engines** (`insightEngine.js`): Slow-Moving Stock, Best Seller Identified, Revenue Growth Milestone, Healthy Repeat Rate) + **6 SQL-based segments**. All references to "8 engines" elsewhere in this doc mean this combined set.

Confirmed existing schema:

- `marketing_campaigns`: id, company_id, opportunity_id, name, type, segment, objective, status, target_count, customers_targeted, customers_converted, expected_impact, actual_revenue, campaign_cost, conversion_rate, roi, notes, ai_content, campaign_snapshot, launched_at, completed_at, created_at
- `marketing_campaign_targets`: id, company_id, campaign_id, customer_id, created_at
- Core overlaps: `customers`, `sales`, `inventory`, `products`, `credits`

Confirmed existing API (`/api/marketing`): `GET /opportunities`, `GET /segments`, `GET /segments/:segmentId/customers`, `GET /campaigns`, `GET /campaigns/:id`, `POST /campaigns/generate`, `PUT /campaigns/:id`, `GET /campaigns/:id/performance`, `GET /campaigns/:id/export/csv`, `GET /dashboard`, `POST /campaigns`, `DELETE /campaigns/:id`.

Signal Engine build plan (approved):

1. `marketing_signals` table — persists engine output (currently computed on-the-fly via SQL, never stored unless a campaign is generated).
2. `knowledge_graph_edges` table — relational approximation of a graph on SQLite: `(node_a, node_b, relationship_type, weight)` (e.g., `customer:123 – bought_during – festival:diwali`). **Approved over a real graph DB (Neo4j) — incompatible with current SQLite/embedded architecture, not worth the new infra dependency at this stage.**
3. **Wrap, don't modify** — the existing `getMarketingOpportunities()` function stays untouched. A cron job (see `node-cron` note below) periodically runs it and inserts results as nodes/edges into the new tables.
4. Downstream Phase 2+ features query `marketing_signals` / `knowledge_graph_edges` directly, never re-run raw engine SQL.

Do not touch: the existing 4+4 engines' core logic. Everything new either consumes their output via the new signal tables or adds a new engine that writes into the same tables.

Phase 1 — Core Marketing Foundation

Feature	Description	Tier	Autonomy	Confirmed Build Details
Customer Segmentation (RFM + Rule-Based)	Auto-buckets by Recency/Frequency/Monetary; custom rule-based segments too.	●	● Full Auto	New tables: <code>custom_segments</code> , <code>segment_rules</code> . New endpoint: <code>POST /segments</code> .

Feature	Description	Tier	Autonomy	Confirmed Build Details
Campaign Builder	Create/schedule campaigns across Email, SMS, WhatsApp, Push.		 Approve	Extends existing <code>marketing_campaigns</code> table + <code>POST /campaigns/generate</code> .
Coupons & Promotions Engine	%/flat/BOGO coupons, segment-targeted, auto-applied, usage limits & expiry.		 Approve above margin floor	New tables: <code>coupons</code> , <code>coupon_redemptions</code> . New endpoints: <code>POST /coupons</code> , <code>GET /coupons</code> , <code>POST /coupons/validate</code> .
Loyalty Points (Earn/Burn)	Points per ₹ spent, redeemable for discounts.		 Full Auto	Merged with Wallet below — one system, not two.
Customer Wallet	Digital wallet holding cashback, reward points, gift cards, and prepaid store credit in one balance.		 Approve for cashback issuance above threshold	New tables: <code>customer_wallets</code> , <code>wallet_transactions</code> . New endpoints: <code>GET /wallet/:customerId</code> , <code>POST /wallet/adjust</code> .
Omnichannel Delivery (WhatsApp, SMS, Email, Push)	Unified send layer across channels, no duplicate messaging.		 Full Auto (execution only)	New table: <code>communication_logs</code> (tracks Send/Open/Click). New endpoints: <code>POST /communications/send</code> , <code>POST /communications/webhook</code> . Build against a mock adapter first — see provider note below.
Post-Purchase Feedback Surveys	Auto-sent star rating + comment after purchase/service.		 Full Auto	New tables: <code>feedback_surveys</code> , <code>survey_responses</code> . New endpoints: <code>POST /surveys/send</code> , <code>POST /surveys/submit</code> .

Feature	Description	Tier	Autonomy	Confirmed Build Details
Basic Campaign Dashboard	Sends, opens, clicks, conversions, revenue-per-campaign.		—	Extends existing <code>GET /dashboard</code> .
Referral Codes (Double-Sided)	Unique shareable code, both sides rewarded.		Full Auto	New tables: <code>referral_codes</code> , <code>referral_uses</code> . New endpoints: <code>GET /referrals/:customerId</code> , <code>POST /referrals/redeem</code> .

Overlap warning (confirmed by Antigravity): several tables above reference `customers` and `sales` — use foreign keys, never duplicate customer data into new tables.

Omnichannel provider decision: build the send layer against an adapter interface (`sendMessage(channel, payload)`), ship with a mock adapter for Phase 1 so everything downstream (campaigns, wallet notifications, surveys) can be built and tested without live API costs. Real provider (MSG91/Gupshup recommended for India — handles DLT registration) gets plugged in later without touching upstream code.

Phase 2 — Owner Spend Intelligence Layer

Feature	Description	Tier	Autonomy
Channel ROI Ranking	Ranks channels by actual ₹-return per ₹-spent.		—
Suggested Budget Split	Proposes allocation across channels/segments by marginal ROI.		Approve
Segment Spend Priority	Ranks which segments are worth spending on this week, from Signal Engine output.		Full Auto (ranking only)
Break-Even Spend Calculator	Minimum conversion rate needed for an offer to not lose money.		—
Rupee-Per-Rupee Framing (UI-wide)	Every number shown as "₹X spent → ₹Y expected back."		—

Feature	Description	Tier	Autonomy
Weekly One-Line Recommendation	Single push/WhatsApp nudge with the week's top spend action.		 Approve (below threshold auto-executes)
Post-Spend Report Card	Plain-language verdict after each campaign.		 Full Auto
Do-Not-Spend Flags	Flags segments/channels with consistently poor ROI.		 Full Auto (advisory)
Store Health Score	One AI-generated score (0-100) summarizing overall marketing effectiveness — spend efficiency, retention trend, engagement, review sentiment — with a ranked list of the top 3 actions to improve it. Becomes the single number an owner checks, the way a credit score is a single number a bank checks.		— (score + advisory actions)

Phase 3 — Automation & Lifecycle

Feature	Description	Tier	Autonomy
Lifecycle Stage Tracking	Auto-classifies: New → Active → At-Risk → Lapsed → Win-Back.		 Full Auto
Win-Back Sequences	Multi-step sequence for lapsed customers, escalating offer value.		 Approve first run, then 
Welcome & Onboarding Journey	Sequence targeting the 2nd/3rd purchase.		 Full Auto
Event Trigger Engine	Unifies every trigger source into one orchestration layer — birthdays, anniversaries, salary days, festivals, weather changes, sports events, local events, inventory changes — so campaigns fire from a single rules engine instead of scattered one-off		 Approve for new trigger types,  once proven

Feature	Description	Tier	Autonomy
	automations. Requires introducing <code>node-cron</code> — current backend is request-driven only, no background worker/scheduler exists yet. Approved as lightweight, no new infra dependency.		
Inventory-Triggered Marketing	Auto-sends "back in stock"/"only 3 left" tied to live inventory. <i>(now a rule inside the Event Trigger Engine)</i>		 Full Auto
Margin-Aware Spend Cap	Blocks/warns when a discount would breach the margin floor.		 Human-Only to override
Dynamic Pricing & Promotion Optimizer	Recommends discount depth per offer based on live demand signals, current inventory levels, seasonality, and margin — instead of the owner picking a fixed % out of habit. Reads directly from the Knowledge Graph (inventory + demand + margin nodes).		 Approve above margin floor,  within it
Subscription & Membership Marketing	Supports recurring plans, paid membership clubs, and renewal campaigns — useful for salons, gyms, cafes, and B2B repeat-supply customers. Renewal reminders and lapsed-membership win-backs run through the same lifecycle engine.		 Approve for renewal offers
Geo-Fencing Campaigns → Pincode-Targeted Campaigns	Pivoted per Antigravity's audit: true GPS geo-fencing requires a persistent background mobile app, which isn't in the stack — physically not executable via WhatsApp/SMS/Email. Replaced with pincode-level targeting using the existing <code>customers</code> billing address field. Delivers most of the hyper-local value with zero new infrastructure. This feature is now merged into Local Area Intelligence (Phase 4) rather than kept separate.		 Approve
Frequency Capping / Fatigue Guard	Caps messages per customer per week across all campaigns.		 Full Auto
Cash Flow-Aware Spend Gating	Won't push a big spend if cash is tight, even with good ROI.		 Full Auto (gate only)

Feature	Description	Tier	Autonomy
Customer Concentration Risk Alert	Flags revenue over-dependence on a small % of customers.	●	● Human-Only
Discount Addiction Warning	Flags segments trained to only buy on discount.	●	● Human-Only

Phase 4 — Predictive & Advanced Analytics

Feature	Description	Tier	Autonomy
Churn Prediction	ML score per customer for return-likelihood within X days.	●	● Full Auto (scoring)
Customer Lifetime Value (CLV)	Calculated + predicted CLV per customer.	●	● Full Auto (scoring)
Cohort & Basket Affinity Analysis	Retention by cohort; products frequently bought together.	●	—
Cross-Sell & Upsell Recommendation Engine	Turns basket affinity into actionable next-best-offer suggestions per customer — bundles, complementary products, upgrade paths — surfaced at checkout, in WhatsApp receipts, or in campaigns.	●	● Approve for new bundle types, ● once proven
Marketing Experiment Lab	Systematically tests offers, pricing, creative variants, audience segments, and send times — not just A/B on one dimension at a time — and builds a persistent knowledge base per business of "what actually works here," which feeds every other engine's confidence score.	●	● Full Auto for test execution, ● Approve to adopt a new winning strategy at scale
Local Area Intelligence	Compares performance by locality, pincode, or branch; recommends hyper-local campaigns where one branch's customers respond very differently from another's. Now also the home for Pincode-	●	● Approve

Feature	Description	Tier	Autonomy
	Targeted Campaigns (the practical replacement for true geo-fencing — see Phase 3), using the existing customers billing address field.		
Customer Journey Visualization	Maps every touchpoint from first visit to repeat purchase — shows exactly where customers drop off (e.g., "many first-time buyers never return for a 2nd purchase within 30 days") so the owner sees the leak, not just the aggregate churn number.	●	— (visualization)
A/B Testing (Message, Offer, Send-Time)	Split-test variants, auto-roll out the statistical winner. <i>(now a component of the Experiment Lab)</i>	●	● Full Auto
Sentiment Analysis on Feedback	AI reads free-text feedback/reviews, extracts themes.	●	—
Confidence Score on Every Recommendation	Every recommendation ships with a confidence % — now computed centrally by the Signal Engine.	●	—
Track Record Scoreboard	Running tally of past recommendations vs. actual outcomes.	●	—
Benchmark Reports (Anonymized Peer Comparison)	Compares this business against anonymized aggregates of similar BizBook MSMEs.	●	—

Phase 5 — India-Context Signal Engine (the moat)

Feature	Data Source	Tier	Autonomy
Festival & Muhurat Calendar Triggers	Diwali/Eid/Holi/Pongal/regional New Years + auspicious dates. <i>(fires via Event Trigger Engine)</i>	●	● Approve
Weather-Triggered Spend Guidance	IMD/OpenWeather forecast tied to footfall/product relevance.	●	● Full Auto

Feature	Data Source	Tier	Autonomy
Monsoon Onset Tracking	IMD regional monsoon onset forecast for seasonal categories.	●	● Approve
Mandi/Commodity Price Feed	Agmarknet — raw material cost swings tied to promotion timing.	●	● Approve
Fuel Price Index	PPAC/data.gov.in — informs delivery-promo margin safety.	●	● Full Auto (gate)
Cricket/IPL Match Calendar Trigger	Public sports schedule — footfall/delivery spikes on match days.	●	● Full Auto
Local Disruption Alerts (Bandh/Strike)	News/transport union notices — suppresses spend on dead days.	●	● Full Auto (suppression)
Government Scheme Radar	data.gov.in / MSME Ministry scheme pages, matched to Udyam category/turnover.	●	● Human-Only
Micro-Influencer & Creator Campaigns	Manages local influencer/creator collaborations — assigns trackable coupon codes per creator, measures redemptions and revenue attributed, calculates true ROI per collaboration instead of guessing based on follower count.	●	● Approve
Competitor Promotion/Size Signal	Public GSTIN lookups + observed local signals.	●	—
Exam/Board Calendar Awareness	CBSE/state board schedules — demand dips/spikes near schools.	●	● Full Auto

Phase 6 — AI Co-Pilot & Autonomous Layer

Feature	Description	Tier	Autonomy
Natural Language "Ask Your Data"	Plain-language question, direct answer with an action button.	●	—

Feature	Description	Tier	Autonomy
Voice-Note Campaign Creation	Voice note → fully configured campaign for review.	●	● Approve
AI Marketing Co-Pilot Chat	Full-context chat that can answer and execute actions.	●	Follows Autonomy Registry
AI Creative Studio	Generates WhatsApp messages, email templates, posters/banners, product images, social posts, and short videos tailored to a specific campaign and the business's brand look — not just copywriting, full creative asset generation.	●	● Approve (visual/public-facing output)
Review & Reputation Management	Pulls reviews in from Google, WhatsApp, and other sources into one place, runs sentiment analysis across all of them, and — critically — identifies which specific happy customers should be asked for a review right now (based on recent positive interaction), rather than blanket-asking everyone.	●	● Full Auto to detect + request, ● Approve to publish any AI-drafted response
Explainable "Why This" Tags	One-line plain-language reason on every recommendation.	●	—
WhatsApp-Native Owner Nudges	Recommendations pushed directly to the owner's phone.	●	● Full Auto (delivery only)
Executive Daily Brief	Every morning: what happened yesterday, what needs attention today, and recommended actions with projected ₹ outcomes — one message, no dashboard visit required. This absorbs and upgrades the earlier Weekly Recommendation into a daily cadence once enough signal volume exists.	●	● Full Auto (delivery), underlying actions follow their own registry entries
Auto-Pause Underperforming Campaigns	Alerts owner mid-flight if a live campaign is underperforming.	●	● Full Auto to pause, ● to resume

Feature	Description	Tier	Autonomy
Autonomous Campaign Agent (opt-in)	Independently launches low-risk campaigns without manual approval.	●	● Full Auto — opt-in only, per Leash Settings
Marketing Knowledge Graph (query surface)	The Co-Pilot's ability to traverse the graph directly — "show me every customer who bought umbrellas last monsoon and hasn't returned since" — is what makes natural-language queries actually powerful instead of shallow keyword search.	●	—

Build Sequencing Summary

Phase	Focus	Depends On
0	Signal Engine + Knowledge Graph foundation	—
1	Core execution layer + Customer Wallet	Phase 0
2	Owner Spend Intelligence + Store Health Score	Phase 1 + Signal Engine
3	Automation, Event Trigger Engine, Dynamic Pricing, Subscriptions, Geo-Fencing	Phase 1 + 2
4	Predictive analytics, Experiment Lab, Local Area Intelligence, Journey Viz, Cross-Sell Engine	Phase 1-3 data volume
5	India-context signals + Micro-Influencer Campaigns	Parallel to Phase 3/4
6	AI Co-Pilot, Creative Studio, Reputation Management, Executive Daily Brief	Phases 2-5 feeding one interface

Recommended pitch order to Antigravity: 0 → 1 → 2 → 3, then run 4 and 5 in parallel (4 is data science, 5 is data engineering/scraping), then 6 last as the synthesis layer. Geo-Fencing should be scoped separately as its own consent/privacy mini-spec before Phase 3 development starts, since it's the one feature here with real regulatory exposure if built carelessly.